# Alexandre S. Avaro

alexandre.avaro@polytechnique.edu ◇ linkedin.com/in/alexandre-avaro
Google Scholar

## EDUCATION

**Ph.D. in Mechanical Engineering**, Stanford University, GPA 4.1/4.0 — Sep 2020 - Aug 2025
Thesis title: CRISPR assays and microfluidic devices for quantitative diagnostic methods.
Supervisors: Juan G. Santiago & Andrew D. Griffiths.

**Master of Science in Mechanical Engineering**, Stanford University, GPA 4.07/4.0 — Sep 2020 - Jun 2022
Depth: Fluid Mechanics. Breadth: Solid Mechanics.

**Diplôme d'ingénieur**, CentraleSupélec (Université Paris-Saclay), GPA 4.0/4.0 — Sep 2018 - Jun 2022
Major: Aeronautics, Space, Transport. Bachelor in Engineering completed in Sep 2019.

**Classes Préparatoires aux Grandes Écoles PCSI-PC***, Lycée Louis le Grand, GPA 4.0/4.0 — 2016 - 2018

## EXPERIENCE

**Postdoctoral Researcher**, École Polytechnique & Institut Pasteur, CNRS — Sep 2025 - Present
LadHyX & Physical Microfluidics and Bioengineering team, under the supervision of Profs. Charles N. Baroud and Gabriel Amselem. *Palaiseau, France*

- Hydrogel-based actuators for mechanobiology.
- Supervised two graduate interns (M. Ratte and S. Rao) and one undergraduate intern (M. Mnif).

**Research Engineer**, ESPCI, CNRS — Jan 2023 - Aug 2025
Biochemistry Laboratory, under the supervision of Prof. Andrew D. Griffiths. *Paris, France*

- Control and dynamics of DNA in multifurcation microfluidic networks using isotachophoresis.
- CRISPR-based diagnostics for trisomy 21. Developed high-throughput sequence screening pipeline for CRISPR Cas12 using droplet-based microfluidics and machine learning.
- Co-supervised one undergraduate intern (C. Dupire).

**Graduate Research Assistant**, Stanford University — Oct 2020 - Aug 2025
Stanford Microfluidics Laboratory, under the supervision of Prof. Juan G. Santiago. *Stanford, CA*

- Developed and experimentally validated isotachophoresis/electrophoresis simulators CAFES and BEAN.
- Developed novel techniques for nucleic acid diagnostics combining CRISPR and microfluidics.
- Studied fundamental theory of CRISPR kinetics and its link to the limit of detection of CRISPR-based assays for molecular diagnostics.
- Supervised three undergraduate interns (L. Agrawal, L. Brown, A. Schwarzbach).

**Teaching Assistant**, Stanford University — Jan 2022 - Jun 2025
ME360: Physics of Microfluidics, taught by Prof. Juan G. Santiago. (graduate level) *Stanford, CA*
ME354: Experimental Methods in Fluid Mechanics, taught by Prof. Juan G. Santiago. (graduate level)
ME351B: Fluid Mechanics, taught by Prof. Juan G. Santiago. (graduate level)
CME102: Ordinary Differential Equations, taught by Prof. Hung Le. (undergraduate level)

**Teaching Assistant**, CentraleSupélec — Feb 2019 - May 2019
Quantum and Statistical Physics, taught by Prof. Jean-Michel Gillet. (undergraduate level) *Gif-sur-Yvette, France*

## PEER-REVIEWED PUBLICATIONS

Qi Jiang, **Alexandre S. Avaro**, Hyunjun Bae, Angelina Sorensen, and Juan G. Santiago, "Experimental Methods for CRISPR Enzyme Assays with Fluorescence Readout", *ACS Sensors*, 11, 10, 8780–8797, **2026**. Link.

Qi Jiang, Ashwin Ramachandran, **Alexandre S. Avaro**, Diego A. Huyke, and Juan G. Santiago, "Reaction kinetics of CRISPR *trans*-cleavage controlled using isotachophoresis", *Analytical Chemistry*, 97, 50, 27646-27653, **2025**. Link.

**Alexandre S. Avaro**, Shahab Mirjalili, Andrew D. Griffiths, and Juan G. Santiago, "Microfluidic networks using isotachophoresis", *Proceedings of the National Academy of Sciences*, 122, 45, e2511724122, **2025**. Link.

**Alexandre S. Avaro** and Juan G. Santiago, "Engineering guidelines for CRISPR diagnostics", *Chemical Communications*, 61, 95, 18756-18772, **2025**. Link. Featured on the inside front cover of *Chemical Communications*.

**Alexandre S. Avaro**, Andrew D. Griffiths, and Juan G. Santiago, "Degradation of Reporter Molecules Imposes a Fundamental Limit of Detection on CRISPR Diagnostics", *Analytical Chemistry*, 97, 25, 13192-13200, **2025**. Link.

Qi Jiang, Xuehao Zang, Yilu Wang, **Alexandre S. Avaro**, Diego A. Huyke, and Juan G. Santiago, "A Three-Dimensional Electrokinetic Microfluidic Device Embedded within a Thermal Cycler Tube", *Lab on a Chip*, 25, 3962-3970, **2025**. Link.

**Alexandre S. Avaro**,† Adar Schwarzbach,† Amit Jangra, Supreet S. Bahga, and Juan G. Santiago, "Highly Parallel Simulation Tool for the Design of Isotachophoresis Experiments", *Analytica Chimica Acta*, 1337, 343553, **2025**. Link. Featured on the outside front cover of *Analytica Chimica Acta*. †: Equal contribution.

Diego A. Huyke,† **Alexandre S. Avaro**,† Thomas Kroll, and Juan G. Santiago, "Stream Lamination and Rapid Mixing in a Microfluidic Jet for X-ray Spectroscopy Studies", *Flow*, 3, E25, **2023**. Link.

**Alexandre S. Avaro** and Juan G. Santiago, "A Critical Review of Microfluidics-Driven CRISPR Diagnostics", *Lab on a Chip*, 23, 938-963, **2023**. Link.

**Alexandre S. Avaro** and Juan G. Santiago, "Uncertainty Quantification of Michaelis–Menten Kinetic Rates and Its Application to the Analysis of CRISPR-Based Diagnostics", *Angewandte Chemie International Edition*, 61, 45, e202209527, **2022**. Link.

Charles Blanluet, Diego A. Huyke, Ashwin Ramachandran, **Alexandre S. Avaro**, and Juan G. Santiago, "Detection and Discrimination of Single Nucleotide Polymorphisms by Quantification of CRISPR-Cas Catalytic Efficiency", *Analytical Chemistry*, 94, 43, 15117–15123, **2022**. Link.

**Alexandre S. Avaro**, Yixiao Sun, Kaiying Jiang, Supreet S. Bahga, and Juan G. Santiago, "Web-Based Open-Source Tool for Isotachophoresis", *Analytical Chemistry*, 93, 47, 15768–15774, **2021**. Link.

## PREPRINTS

**Alexandre S. Avaro**, Leon V. Gebhard, Gabriel Amselem, and Charles N. Baroud, "Zero-Mean Oscillations Drive Recirculations in Potential Flows at Vanishing Reynolds Number", *arXiv*:2609.31196, **2026**. Link.

Leon V. Gebhard, **Alexandre S. Avaro**, Gabriel Amselem, and Charles N. Baroud, "Designing single-layer PDMS devices for micron to millimeter-scale deformations", *arXiv*:2605.17402, **2026**. Link.

## CONFERENCES

### Contributed talks

**Alexandre S. Avaro**, Leon V. Gebhard, Gabriel Amselem, and Charles Baroud, "Multi-material 4D printing of pre-programmed 3D microstructures", *NanoBioTech*, Montreux, Switzerland, **2026**. Link.

**Alexandre S. Avaro**, Leon V. Gebhard, Gabriel Amselem, and Charles Baroud, "Programming flows without inertia: Net recirculations from reciprocal deformations", *Graphyz3*, Marseille, France, **2026**. Link.

**Alexandre S. Avaro**, Leon V. Gebhard, Gabriel Amselem, and Charles Baroud, "Flow patterning at near-zero Reynolds number", *Journée Écoulements Fluides Saclay*, Orsay, France, **2026**. Link.

**Alexandre S. Avaro**, Leon V. Gebhard, Gabriel Amselem, and Charles Baroud, "Locally recirculating flow at near-zero Reynolds number", *Microfluidic Horizons*, Padua, Italy, **2026**. Link.

**Poster presentations**

**Alexandre S. Avaro**, Andrew D. Griffiths, and Juan G. Santiago, "Electrokinetic Device for Parallel Enzymatic DNA Amplification Assays Compatible with Raw Input Sample", *The 28$^{th}$ International Conference on Miniaturized Systems for Chemistry and Life Sciences - Micro-Total Analysis Systems (μTAS)*, Montréal, Canada, **2024**. Link. Awarded CBMS travel grant.

**Alexandre S. Avaro**, Andrew D. Griffiths, and Juan G. Santiago, "Electric-field driven microfluidic devices for affinity-based protein separation", *Single-Molecule Sensors and NanoSystems International Conference*, Barcelona, Spain, **2023**. Link.

**Alexandre S. Avaro** and Juan G. Santiago, "Uncertainty Quantification of Michaelis–Menten Kinetic Rates and Its Application to the Analysis of CRISPR-Based Diagnostics", *Microfluidics Global Exposition*, Paris, France, **2022**.

## PATENTS

**Alexandre S. Avaro** and Juan G. Santiago. "A microfluidic chip for automated, parallel nucleic acid testing from raw samples". Application. Link.

**Alexandre S. Avaro** and Juan G. Santiago. "Apparatuses and Methods involving Spatiotemporal Sequencing of Droplet Assays". Application WO2024148025A1 (patent pending).

Juan G. Santiago, **Alexandre S. Avaro**, Ashwin Ramachandran, Diego A. Huyke, and Charles Blanluet. "Enzyme Kinetics Analyses of CRISPR Endonucleases". Application WO2023201275A1 (patent pending, licensed).

## SKILLS

**Experimental skills.** Microfluidic system design and control, microfabrication and prototyping (PDMS chip fabrication, soft lithography, 3D printing using photolithography, CNC machining), epifluorescence microscopy, quantitative image processing, electrokinetic system control, biomolecular techniques (sample preparation, molecular biology experimental design: PCR, qPCR, LAMP, IVT, CRISPR-Cas12), data analysis and processing.

**Software.** Python, MATLAB, Mathematica, Inkscape, Microsoft Office Suite, and LaTeX. Basic proficiency with JavaScript (React), keras, and Adobe InDesign.

**CAD and simulation software.** Proficient with COMSOL Multiphysics, SOLIDWORKS, AutoCAD, Fusion360, and Rhino/Grasshopper.

**Languages.** French (native), English (fluent, TOEFL iBT 112/120, IELTS 8.0), German (B2+ CEFR).

## SERVICE

**Peer reviewer for archived journals.** *Sensors & Diagnostics* (2026), *ACS Omega* (2026), *Analytica Chimica Acta* (2025), *Lab on a Chip* (2025).

**Competition judge.** iGEM (international competition in molecular biology with 400+ undergraduate and postgraduate teams), 2023 and 2024.

**Science outreach.** Participated in science popularization events, including Nuit Européenne des Chercheur·es (European Researchers' Night) in ESPCI Paris.

**Student representative in CentraleSupélec.** Participated in the design of the new curriculum of CentraleSupélec.

**Research infrastructure.** Built a database of mechanical test devices in CNRS laboratories in the Paris area, supervised by Dr. Eva Héripré (CNRS).